
An AI Poker Agent for 2-Player Limit Texas Hold'em: Search, Equilibrium Approximation, and Evolutionary Hybrids

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Abstract

We construct and evaluate four independent agents for 2-player limit Texas Hold'em, each attacking the game's $\sim 10^{17}$ -state imperfect-information tree from a different angle: pure Monte Carlo statistical rollouts (**PUREMCAGENT**), Monte Carlo Counterfactual Regret Minimization with a Bayesian exploit layer (**MCCFRAGENT**), a depth-3 minimax search with a learned linear leaf evaluator and semi-bluff logic (**HYBRIDAGENT**), and a GA-tuned extension with persistent opponent modeling and a three-tier bluff pipeline (**GAUNTLETAGENT**). A two-stage selection protocol picks a single agent for the class tournament based on (i) average leaderboard gain across the cohort and (ii) a controlled head-to-head under a fixed deal sequence so that variance from card luck is eliminated. We submit **GAUNTLETAGENT**, whose GA-tuned linear evaluator and profile-weighted opponent model give the most balanced performance against an unknown opponent distribution, achieving the best average cross-cohort edge (+7.14 chips/scenario) and the smallest worst-case loss (−1.89) across all internal matchups.

1 Introduction

Two-player limit Texas Hold'em has roughly 10^{17} game-tree nodes and was only weakly solved at near-equilibrium accuracy in 2015 by Bowling et al. [4]. Brute-force search is infeasible: the game has stochastic deals, hidden information, and a branching factor that explodes with each betting round. Every practical agent therefore combines three ingredients in some form: *abstraction* of the state space, *simulation* or *equilibrium computation* for value estimates, and an *opponent model* for irrationality and asymmetry. Our group built four agents that each emphasise a different combination of these ingredients and ran a tournament-style selection protocol to choose a single submission.

1.1 Our contributions

We construct four agents and a controlled selection protocol over them. **PUREMCAGENT** (Martin Riddell) is a guided random-rollout agent backed by a 265M-simulation pre-flop equity table and time-bounded post-flop Monte Carlo. **MCCFRAGENT** (Nhi Ha) is a 2M-iteration external-sampling MCCFR strategy table over an 8-bucket card abstraction, augmented at runtime with a Bayesian opponent-exploit layer. **HYBRIDAGENT** (Aarti Kumari) is a depth-3 α - β minimax search with a 14-feature linear leaf evaluator and deterministic semi-bluff and slow-play triggers, with weights trained offline via Q-Learning. **GAUNTLETAGENT** (Dhevdharsan Bhavani Satish Kumar) extends **HYBRIDAGENT** with 50-simulation MC hand-strength estimation, a three-tier bluff pipeline, persistent cross-round opponent modeling, and weights evolved by a Genetic Algorithm on a 4-opponent Gauntlet. The selection protocol uses two criteria: average leaderboard gain across the class cohort, and a fixed-deal head-to-head among the four entries.

1.2 Related work

Counterfactual Regret Minimization [8] provides the theoretical scaffold for MCCFRAGENT: in two-player zero-sum games, CFR iteratively minimises regret and provably converges to a Nash equilibrium. The external-sampling Monte Carlo variant is due to Lanctot et al. [7] and Lanctot [6], and the same family of techniques scaled to professional play in LIBRATUS and PLURIBUS [5]. Opponent modelling in poker dates back to Billings et al. [3], who showed that profile-based action prediction yields stronger adaptive play than static strategies; we use lightweight versions of this idea in both MCCFRAGENT’s exploit layer and GAUNTLETAGENT’s profile-weighted minimax. Our use of a Genetic Algorithm to learn the linear leaf evaluator follows Angeline et al. [1], who applied population-based search to game-playing agents as an alternative to gradient methods.

2 Model and preliminaries

We model the game in standard limit hold’em form: two players, four betting streets, and three legal actions per turn (fold / call / raise). The hand bucket $b(\vec{h}, \vec{b}, t) \in \{0, \dots, 7\}$ at street t is computed by a Chen-style score preflop and a hand-category mapping postflop [6]; it is the primary abstraction shared across agents.

PUREMCAGENT (rollout-based). A pure simulation agent with no game-tree search and no opponent model. It applies a threshold rule on an estimated win probability $\hat{p}$ with cutoffs set at the raise/call and call/fold boundaries of the empirical preflop equity distribution. Pre-flop $\hat{p}$ comes from a memoization table of 2,652 ordered hole-card pairs, each estimated with 10^5 simulations; post-flop $\hat{p}$ is re-estimated by random rollouts inside a 0.10 s time budget. See Section A.1 for full implementation details and a discussion of the scalability limitations of this approach.

MCCFRAGENT (equilibrium-based). A pre-trained MCCFR strategy table loaded at runtime, with information sets keyed by (t, p, b, a_t) : street, position, bucket, and on-street action history. Trained for 2×10^6 iterations on 140 active info-sets; the average policy entropy drops from 0.424 to 0.374 over training, indicating convergence toward equilibrium. A runtime Bayesian layer shifts the base policy when the opponent’s empirical fold or raise rate exceeds calibrated thresholds. See Section A.2 for more details.

HYBRIDAGENT (Minimax + Alpha-Beta with Semi-Bluff). Depth-3 α - β minimax over an abstract state space, with a 14-feature linear leaf evaluator $\phi(s) = \sum_{j=1}^{14} w_j \phi_j(s)$. At each opponent node the search uses a profile-weighted expected value over {aggressive, passive, balanced}, classified from observed action frequencies. Two deterministic bluff triggers short-circuit the search on a small fraction of eligible states: a semi-bluff raise on flush/straight draws ($\sim 25\%$ of eligible flop/turn situations) and a slow-play call on monster hands ($\sim 15\%$ of eligible preflop/flop situations). Weights were trained offline via Q-Learning. See Section A.3 for more details.

GAUNTLETAGENT (GA-Trained Adversarial Search). Extends HYBRIDAGENT with three improvements: (i) 50-simulation Monte Carlo hand-strength estimation postflop in place of the closed-form heuristic; (ii) a three-tier bluff pipeline adding a pure-river-bluff trigger and a calling-station filter that disables bluffing against opponents who never fold; and (iii) weights evolved by a Genetic Algorithm on a 4-opponent Gauntlet (PUREMCAGENT, RANDOMPLAYER, STRONGRULEPLAYER, plus a co-evolutionary sibling), producing a balanced tight-aggressive (TAG) generalist policy. See Section A.4 for more details.

3 Results

3.1 Experimental setup

All experiments use the standard pypokerengine runner with two players, an initial stack of 1,000 chips, a small blind of 10, a big blind of 20, and a maximum of 10 raises per street. Two evaluations feed the selection protocol: a class-wide leaderboard run exposing each agent to a heterogeneous opponent cohort, and a controlled head-to-head between every pair of our four agents under a fixed deal sequence so that card-luck variance is eliminated and any margin reflects pure decision quality.

3.2 Selection process

We compare the four agents under two complementary criteria. **Criterion 1 (leaderboard gain)** measures performance against the broad class cohort, a robust proxy for expected performance against an unknown opponent. **Criterion 2 (fixed-deal head-to-head)** measures internal performance within our own pool under a controlled deal sequence (Table 1).

Winner	Loser	Edge	N	Win%
GAUNTLETAGENT	MCCFRAGENT	+16.16	2,000	57.6
HYBRIDAGENT	GAUNTLETAGENT	+1.89	6,000	52.0*
HYBRIDAGENT	MCCFRAGENT	+2.32	4,000	46.6
PUREMCAGENT	HYBRIDAGENT	+13.54	2,000	41.1
MCCFRAGENT	PUREMCAGENT	+3.21	2,000	34.7 [†]

Table 1: Fixed-deal head-to-head results. Edge = avg chips/scenario (row-agent perspective). *Partial run. [†]Win-count near-tie; edge favours MCCFRAGENT in chip magnitude.

The four agents form a complete non-transitive cycle, a rock-paper-scissors structure [2] in which no agent dominates across all opponent styles:

GAUNTLETAGENT $\succ$ MCCFRAGENT $\succ$ PUREMCAGENT $\succ$ HYBRIDAGENT $\succ$ GAUNTLETAGENT.

Tournament performance therefore depends critically on the opponent distribution. Table 2 summarises the selection outcome.

Agent	Avg edge	Min edge	Robust?
GAUNTLETAGENT	+7.14	−1.89	✓
PUREMCAGENT	+5.17	−3.21	✓
HYBRIDAGENT	−3.11	−13.54	×
MCCFRAGENT	−5.09	−16.16	×

Table 2: Tournament selection: average and worst-case edge per agent across all matchups. GAUNTLETAGENT achieves the best average edge and the smallest worst-case loss.

Selected agent. HYBRIDAGENT’s head-to-head edge over GAUNTLETAGENT (+1.89) arises because GAUNTLETAGENT’s large negative pressure weight makes it susceptible to marginal raises collecting fold equity. However, this exploit is specific to HYBRIDAGENT’s architecture and is unlikely to generalise to the full leaderboard cohort. GAUNTLETAGENT, trained on a 4-opponent Gauntlet for robustness, tops the leaderboard and achieves the best average edge and smallest worst-case loss across all internal matchups. We therefore submit **GAUNTLETAGENT** as our class tournament entry.

3.3 Why GAUNTLETAGENT wins

Three properties jointly explain the result (Figure 1). First, the GA Gauntlet tuned weights across four opponent classes simultaneously, producing a balanced TAG profile; `hand_strength` rose +84%, `pressure_bucket` grew 4× more negative, and `pair_bonus` rose +162% (Table 4). Second, opponent nodes use profile-weighted expected value rather than pure min, responding differently to aggressive, passive, balanced, and calling-station profiles. Third, the three-tier bluff pipeline (~25% semi-bluffs, ~15% slow-plays, ~20% river bluffs) fires under deterministic state hashes, keeping deception reproducible. Full architectural details are in Section A.

3.4 Discussion

Why GA over RL? Policy gradient RL updates a single weight vector along one gradient direction and is prone to non-convex local optima. The GA maintains 8 diverse individuals and recombines their best features via crossover, escaping basins that gradient methods cannot leave. The empirical result, convergence to a qualitatively different TAG weight profile that wins more chips, validates this hypothesis concretely.

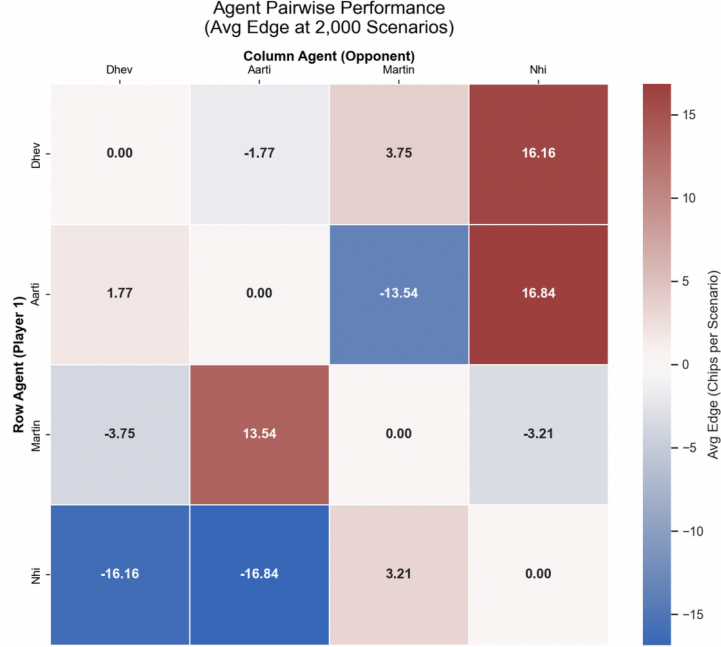


Figure 1: Pairwise performance matrix: average chip edge of the row agent against the column agent over fixed-deal scenarios.

Calling-station detection. The calling-station filter costs only a memory counter and a comparison, and pays for itself within the first 20 rounds. Spending bluff chips on a player who never folds leaks EV at every opportunity; this behavioural adaptation is absent from theory-driven agents such as MCCFRAGENT.

Limitations. GAUNTLETAGENT’s opponent model assumes a stationary profile; an adversary who deliberately shifts style mid-match could degrade the classifier. A change-point detector or Bayesian mixture over profiles would address this. The linear evaluation function also cannot capture interaction effects (e.g., strong hand and high pressure) that a shallow neural network leaf evaluator would handle naturally.

References

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A Detailed agent specifications

A.1 PUREMCAGENT (rollout-based)

The decision rule is a single threshold on an estimated win probability:

$$\pi(\hat{p}) = \begin{cases} \text{raise} & \hat{p} \geq 0.50 \\ \text{call} & 0.20 \leq \hat{p} < 0.50 \\ \text{fold} & \hat{p} < 0.20. \end{cases} \quad (1)$$

Pre-flop $\hat{p}$ is read from a memoization table of 2,652 ordered hole-card pairs, each estimated with 10^5 Monte Carlo simulations ($\approx 2.65 \times 10^8$ total simulations offline). Post-flop, $\hat{p}$ is re-estimated at decision time by random rollouts inside a strict 0.10 s time bound. The empirical equity distribution is unimodal around 0.45: most hands fall in the call region, only premium pairs and high-suited connectors clear the raise threshold of 0.50, and only weak unsuited low-card holdings drop below the fold threshold of 0.20.

Weakness of pure MC Under the .5-seconds-per move time constraint, the pure MC agent was able to simulate several thousand games to obtain a reliable estimate of hand strength with low variance. However, extending this approach to also account for opponent hands introduced a fundamental scalability problem. There was enough time to simulate 1 hand accurately, but simulating all 2,652 possible hands meant that each hand received significantly less time for the rollout, causing the simulated win rates to diverge significantly from the true win rates. Further analysis on this divergence can be found in Section C.4. We considered bucketing to reduce the number of opponent hands needed to simulate, but determined that the estimation errors introduced by this technique was not worth it compared to the possible benefits.

A.2 MCCFRAGENT (equilibrium-based)

Each information set is keyed by the tuple $I = (t, p, b(\vec{h}, \vec{b}, t), a_t)$, where $t \in \{0, 1, 2, 3\}$ is the street index, $p \in \{0, 1\}$ is position, $b \in \{0, \dots, 7\}$ is the hand bucket, and a_t is the action history. We trained MCCFR on 140 active information sets for 2×10^6 iterations. Two diagnostics confirm convergence: the average policy entropy drops monotonically from 0.424 to 0.374, and the average action frequencies stabilise at $(P_{\text{fold}}, P_{\text{call}}, P_{\text{raise}}) \approx (0.16, 0.55, 0.28)$ with no visible drift in the second half of training. The runtime Bayesian exploit layer shifts the policy once $n \geq 15$ voluntary opponent actions have been observed.

A.3 HYBRIDAGENT (Minimax + Alpha-Beta with Semi-Bluff)

Search. At our node we take a max; at the opponent node we take a profile-weighted expected value, not a pure min, because the opponent’s hole cards are hidden and real opponents do not always play optimally. Given a profile $\rho \in \{\text{aggressive}, \text{passive}, \text{balanced}\}$ inferred from action history, we compute

$$V_{\text{opp}}(s) = \sum_{a \in \mathcal{A}} \Pr(a \mid \rho) \cdot V(s, a), \quad (2)$$

with $\Pr(\text{raise}, \text{call}, \text{fold} \mid \text{aggressive}) = (0.55, 0.30, 0.15)$, and analogous distributions for passive $(0.15, 0.55, 0.30)$ and balanced $(0.33, 0.44, 0.23)$ profiles. The search is bounded to depth 3 and pruned with α - β .

Linear leaf evaluator. At search cutoffs: $\phi(s) = \sum_{j=1}^{14} w_j \phi_j(s)$, where the eight base features are hand strength, normalised pot bucket, normalised pressure bucket, opponent aggression rate, normalised street index, and three boolean hole-card features (pair / suited / connected). Six conditional terms activate based on the most recent self-action: strong-raise bonus, semi-bluff bonus, weak-raise penalty, slow-play bonus, weak-call penalty, and fold bias. Weights were trained offline via Q-Learning.

Bluff layer. A semi-bluff raise fires on $\sim 25\%$ of eligible flop/turn situations when a flush or straight draw is present and the opponent is not aggressive ($\sim 35\%$ equity to improve by river justifies the raise). A reverse-bluff slow-play fires on $\sim 15\%$ of eligible preflop/flop situations when hand strength exceeds 0.85, the pot is small, and the opponent is aggressive. Both triggers use a deterministic hash of the round state for full reproducibility.

A.4 GAUNTLETAGENT (GA-Trained Adversarial Search)

Three changes extend the HYBRIDAGENT architecture.

Monte Carlo hand-strength estimation. The fast closed-form heuristic is replaced postflop by a 50-simulation call to `estimate_hole_card_win_rate`, giving a sharper estimate of $\hat{p}$ once community cards are visible. A lookup table of premium preflop equities (AA, KK, $\dots$, KQo) is consulted first to keep top hands precise without simulation overhead.

Three-tier bluff pipeline. Before minimax, three gates are checked in order: (1) semi-bluff on draws ($p_{\text{semi}} < 0.25$, non-aggressive opponent); (2) slow-play on monsters ($h > 0.85$, aggressive opponent, $p_{\text{slow}} < 0.15$); (3) pure river bluff ($h < 0.35$, passive/balanced opponent, opponent checked, $p_{\text{bluff}} < 0.20$). A trash-hand filter instantly folds sub-0.40 hands with no draw potential, saving minimax overhead for decisions that require lookahead.

GA training and the 4-opponent Gauntlet. An initial Q-Learning run produced a brittle policy that overfit a single benchmark. We replaced it with a Genetic Algorithm ($N = 8$ individuals) whose fitness was evaluated on a 4-opponent Gauntlet: PUREMCAGENT, RANDOMPLAYER, STRONGRULEPLAYER, and a co-evolutionary sibling. Each generation: retain top 3, breed 5 children via uniform crossover, mutate with $p_{\text{mut}} = 0.30$. The Gauntlet drove the fold-bias negative and amplified the hand-strength multiplier, producing the TAG profile in Table 4.

B Generalist–specialist tradeoff

The four agents reveal a fundamental tension between generalist stability and specialist dominance. PUREMCAGENT and MCCFRAGENT are generalists by architecture. GAUNTLETAGENT sits in the middle by training: the Gauntlet explicitly balanced four sub-objectives at once. HYBRIDAGENT is inadvertently a specialist: its Q-Learning weights were tuned against a narrow benchmark, making it effective against GAUNTLETAGENT’s pressure response but brittle elsewhere. The two-criteria selection protocol is the correct framing for a class tournament: Criterion 1 captures the unknown-cohort regime the submission will face, while Criterion 2 verifies internal consistency. GAUNTLETAGENT satisfies both.

C Hyperparameters, weights, and reproducibility

C.1 Offline training budgets

Agent	Offline computation	Stored object
PUREMCAGENT	$2,652 \times 10^5 \approx 2.65 \times 10^8$ pre-flop sims	<code>memoization_table.json</code>
MCCFRAGENT	2×10^6 MCCFR iterations	<code>mccfr_strategy.pkl</code> (140 info-sets)
HYBRIDAGENT	Q-Learning, single benchmark	<code>hybrid_weights.json</code> (14 weights)
GAUNTLETAGENT	10-gen GA, 4-opponent Gauntlet	<code>hybrid_weights.json</code> (14 weights)

Table 3: Offline training budgets and stored artefacts for each agent.

C.2 Final weights and GA convergence

Table 4 shows the GA-evolved weights alongside the default Q-Learning baseline. The GA converged to a clear TAG profile: `hand_strength` rose +84%, `pressure_bucket` grew $4\times$ more negative,

and `pair_bonus` rose +162%. The five conditional bonus/penalty terms were frozen throughout to prevent the GA from unlearning foundational poker logic.

Feature	HYBRIDAGENT	GAUNTLETAGENT	Change
<code>hand_strength</code>	+4.80	+8.82	↑ +84%
<code>pot_bucket</code>	+0.90	+3.96	↑
<code>pressure_bucket</code>	-1.00	-4.00	↓ 4× more negative
<code>opp_aggression</code>	-1.10	-1.99	↓
<code>street_progress</code>	+0.50	+0.48	n/a
<code>pair_bonus</code>	+1.50	+3.93	↑ +162%
<code>suit_bonus</code>	+0.60	+0.37	n/a
<code>connected_bonus</code>	+0.40	+0.75	n/a
<code>fold_bias</code>	-0.15	-0.77	↓
<code>strong_raise_bonus</code>	+1.10	+1.10	frozen
<code>semi_bluff_bonus</code>	+0.80	+0.80	frozen
<code>weak_raise_penalty</code>	-2.00	-2.00	frozen
<code>slow_play_bonus</code>	+1.40	+1.40	frozen
<code>weak_call_penalty</code>	-1.40	-1.40	frozen

Table 4: HYBRIDAGENT (Q-Learning) vs. GAUNTLETAGENT (GA Gauntlet) weight comparison. Frozen terms preserve foundational poker logic.

C.3 Reproducibility

All four agents are deterministic given a fixed seed except for the time-bounded rollouts in PUREM-CAGENT and GAUNTLETAGENT’s post-flop MC estimator, and the `explicit random.random()` sampling step in MCCFRAGENT’s policy execution. The two minimax agents are fully deterministic: all bluff triggers use a deterministic hash of the round state, and ties in the minimax score are broken by a fixed priority `raise > call > fold`.

C.4 Divergence from MC simulation

To quantify the estimation error introduced by reduced simulation time, we ran 10,000 randomly sampled hands at time budgets of 0.01s, 0.05s, and 0.1s, measuring mean absolute error (MAE) against a ground truth of 0.45s MC rollouts. As shown in Figure 2, MAE increases from approximately 0.92% at 0.1s to 2.65% at 0.01s, following a near log-linear relationship. The 95% confidence intervals are extremely tight across all budgets, confirming that these MAE estimates are stable and not artifacts of sampling noise. This demonstrates that while a 0.45s budget yields reliable win rate estimates, reducing the budget to accommodate opponent hand enumeration would introduce estimation errors large enough to meaningfully affect decision-making. It is not feasible to use this approach for anything other than precomputing the preflop stage, where there are only 2,652 possible holdings.

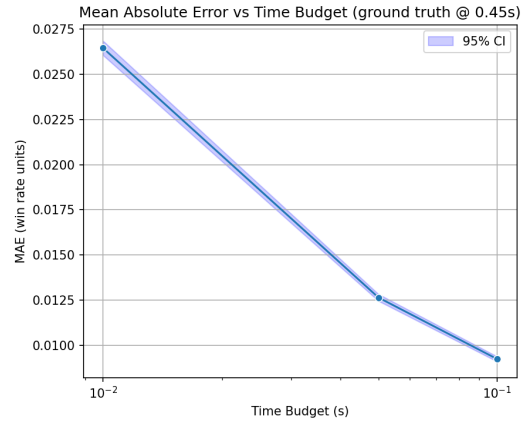


Figure 2: Plotting mean absolute error against the amount of time used for simulations. Simulations were compared to a ground truth of MC rollouts for .45 seconds. A confidence interval of 95% is also shown.